Code Challenge - Full Stack Developer

Intro

The main objective of this challenge is to evaluate your ability to build a complete application, from the data layer (persistence) and business logic (backend) to the user interface (frontend). The emphasis is placed on the implementation of specific functional requirements, namely managing a catalog of courses (for dogs), a user profile, the ability to enroll in a course, mark course steps as complete, and a progression system that unlocks subsequent courses upon completion of the previous ones. Additionally, the system must ensure that courses have a limited number of slots and that enrollment prevents *race conditions* in the number of slots, guaranteeing data integrity and a fluid user experience.

You are tasked with building a small application based on the following functional requirements.

Functional Requirements (User Stories)

1. **Course Catalog:** The user must be able to view available courses, including the price and the number of **remaining seats**.
2. **Enrollment:** The user can enroll in a course.
    - *Golden Rule:* The system must ensure there is no "overbooking". If only 1 seat remains and 2 people click at the same time, only one can succeed.
3. **Blocked Progression:** The user can only view/access "Course B" if they have marked "Course A" as **Completed**.
4. **Profile Overview:** A space to view my enrolled courses and mark a course as finished.

Technical Requirements and Constraints

Backend

- **Language/Framework:** NodeJS
- **Database:** Use a relational database (e.g., PostgreSQL, MySQL)
- **API:** Implement a RESTful API to serve the front-end.
- **Notes:** API authentication is not required

Frontend

- **Framework:** Choose any modern front-end framework (e.g., React, Vue.js).
- **UI/UX:** Focus on clarity and usability. A basic, clean interface is sufficient.
- **Notes:** Platform sign-in and sign-up is not required

Deliverables

- **Source Code:** Provide the complete source code, ideally in a Git repository.

Time Estimate

This challenge is designed to take approximately **8-12 hours** of focused work. We value quality and clean code over rushing to finish.

UI/UX

Feel free to make your own app design or take the images shown below as inspiration. If choosing to go with the design below, there's no need to accommodate for everything, just do what it feels like it makes sense for this challenge.

Hi John!

Go home

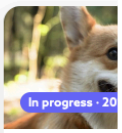
Learn about anything dog related with us.

Available credits

25 credits

Ongoing courses

Potty training
Behaviour

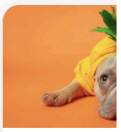


Sitting
Training

Available courses

All

Sed ut alterum aspernandum
Behaviour
costs 1 | 12 slots available



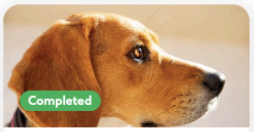
At vero eos e
Nutrition

All categories

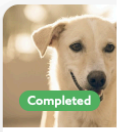
Behaviour

Training

Completed



Sed ut alterum aspernandum
Behaviour



At vero eos e
Nutrition

Future courses

Sed ut alterum aspernandum
Unlock in 1 order

At vero eos e
Unlock in 1 ord

Dashboard | Extras | Vet | Courses | More

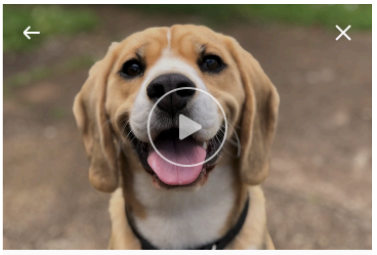
Potty training

At magnum periculum adiit in quo pertineant non ero tibique, si ob aliquam causam ista, quae sunt in gravissimo bello animadversionis metu degendae.

Steps

1. At magnum periculum adiit in quo pertineant non? ✓
2. Sunt autem nusquam hoc epicurus in armatum hostem. ✓
3. In oculis quidem faciunt, ut ratione voluptatem accusantium? ›
4. Hanc ego cum a philosophis compluribus permulta dicantur! ›

Start step 3 →



Step 3

In oculis quidem faciunt, ut ratione voluptatem accusantium?

Torquatos nostros? quos tu tam crudelis fuisse, nihil ut alterum aspernandum sentiamus alii autem, quibus ego cum memoriter, tum etiam erga nos amice et dolore magnam aliquam quaerat voluptatem ipsam voluptatem, quia voluptas nulla pariatur? at vero eos censes tantas res gessisse sine metu contineret.

These treats will help you train your dog by positively reinforcing any good behaviour.

Finish step ✓